

Real-time video anomaly detection — 11 classes, open-set

One frozen encoder, encoded once, read three ways. No backbone fine-tuning.

58.8

benchmark score

18–58×

realtime

125 s

all 28 videos

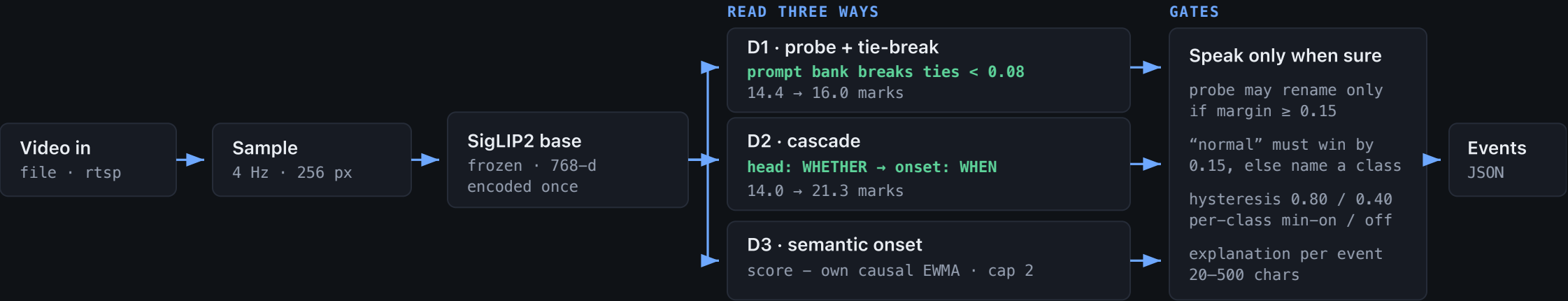
2.4 MB

trained weights

A2000

12 GB GPU

PIPELINE



WHY THIS SHAPE

- **Frozen encoder, one pass.** Features computed once, read three ways — why it beats realtime on a modest GPU.
- **Prompts are data, not weights.** A twelfth class is a sentence, not a retrain.
- **Score against normal.** Margin over the best of ~70 normal prompts; raw similarity just describes the scene.
- **Long video needs a baseline.** D3 keys on the rise over a causal 20 s EWMA, so a busy junction isn't permanently anomalous.
- **Locating ≠ naming.** Detector says when, probe says what — 63% vs 33% on class.

TESTED AND REJECTED

TRIED	OUTCOME
Half-max boundary refinement	D3 16.9 → 8.0
Adaptive per-video D2 thresholds	D2 14.0 → 5.3

Every gain came from routing, not from a bigger model

Three changes moved the score. None added a parameter or a training step — ~30 lines total.

D2: TWO COMPONENTS, EACH DOING HALF THE JOB

	E024 (NORMAL)	E021-23	D2
head only	silent 8.75	5.25	14.0
onset only	fires 0	12.6	12.6
cascade	silent 8.75	12.6	21.3

```
if head.max() >= 0.80: # head answers WHETHER
    found = to_events(...) # onset answers WHEN
```

The head is a good detector and a poor localiser — 0/12 events matched. The onset path is the reverse: it localises well but has no notion of “nothing here”, so it fired on the one normal video and forfeited its marks.

THE SAME IDEA TWICE MORE, ON D1

```
if top == "normal" and p1 - p2 < 0.15: name the anomaly
```

A clip read **normal** 0.132 vs **smoke** 0.131 — a coin flip resolved as silence, forfeiting a whole video. **46.4% → 57.4%**.

```
if p1 - p2 < 0.08: let the prompt bank pick between those two
```

It flipped **fire**↔**smoke** on two videos — the exact pair we had flagged as swapped **by eye**, hours earlier, with no knowledge of any score. **57.4% → 64.1%**.

WHAT ACTUALLY FOUND THE BUGS

- **Looking at the frames.** Rendered the frame at the midpoint of every predicted event into one labelled sheet. Ten minutes, four real defects.
- A 240 s clip called `wrong_way_driving` six times — it was queued trucks on the far carriageway.
- A dashcam **crash compilation** typed `stalled_vehicle` — the “CAR CRASHES TIME” watermark was in frame.
- A correct `fire` call silently destroyed by a later change — caught only by re-rendering.

LOCAL VALIDATION NEVER PREDICTED THE SCORE

LOCAL SIGNAL	SAID	REALITY
Marks formula	29.9	51.9
Train accuracy	+27	-24